



Quasar peering

Group of researchers peered inside the brightest Quasar using ALMA and found out its ready to spill over. <http://dgit.in/ALMApeer>



Chrome Pickup

Chrome pickups new security trends from the spreading Windows 10 to safeguard its users in the new update. <http://dgit.in/ChroPick>

THE NEW SPACE RACE

The idea of deploying satellite clusters into space to bring highspeed internet to virtually anywhere on earth is fine enough. But the challenges seem to be many.

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While space has been something that has fascinated humans since time immemorial, advances in space science has been a relatively recent development in the human history. It was only in the previous century that man managed to break through to space physically and the farther reaches of the heavens still seems to be a long way off.

But that's not to say that human endeavour related to outer space is not progressing. In fact, even at this very moment, a new space race is on, and unlike the race for the conquest of space in our past, this time around it's not nations that are the key competitors but business entities. Also, it's not for exploring outer space that the new race is underway but to make something happen down here on Earth.

And what they aim to do is certainly something that'll benefit earth's population: to provide access to broadband internet for the billions who currently do not have the access which many of us take for granted.

Ways to make things out yonder work for the earthlings

The idea is as simple as its execution is complex: to launch mega satellite clusters into outer space that will provide superfast internet almost anywhere on our planet.



Bezo's Blue Origin-World's first reusable rocket

One of the main terms related to this method is backhaul. Backhaul means the links that have to be established between the satellites and also between the satellites and the internet so that customers can access the internet. This could be made possible using microwave beams or laser beams that function at millimeter wavelengths. This will also require self-aligning systems that can locate other satellites and maintain links even in the case of changes in their relative positions.

Another way in which backhaul can be formed for the satellites is by connecting them to ground stations that are connected to major internet gateways.

The key players racing for the pie in the sky

Though multiple companies are involved in this new space race, two of the major firms involved are OneWeb and SpaceX. Though from two different companies-